

One Generator, Any Scanner: AdaIN-Conditioned Self-Attention Harmonization Across N MRI Sites with Zero-Shot Transfer to Held-Out Institutions

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Abstract. Multi-site MRI studies span many scanners and acquisition protocols, and pairwise generative harmonizers need $O(N^2)$ models to cover N sites. We learn a *single* generator that translates any input to any of N acquisition domains, conditioned through Adaptive Instance Normalization (AdaIN) on a learned per-site style and supervised by a StarGAN-style auxiliary domain classifier, on a self-attention 2.5D tri-planar backbone. We frame cross-scanner discrepancy as a channel-wise intensity-distribution shift that AdaIN site-conditioning is purpose-built to correct, with self-attention capturing its spatially non-stationary component. We instantiate $N=4$ on *real* contributing institutions—four tissue-source sites of TCGA-GBM—and evaluate with an $N \times N$ translation matrix, per-pair Fréchet distance, and a learned site classifier. The single model reduces within-site Fréchet distance to roughly half the cross-site value, drives the site classifier from 0.43 toward its 0.25 chance level, and—most notably—transfers *zero-shot* to three held-out institutions at a fidelity matching its trained cross-site translations (132 vs. 127), evidence that the learned site-style space generalizes beyond the training sites. We report these site distributions to be subtle (a strong real-image classifier reaches only 0.43), making this a conservative testbed, and pair the result with an honest account of where single-model harmonization helps and where its margins are thin.

Keywords: MRI harmonization · multi-domain translation · AdaIN · self-attention · zero-shot transfer.

1 Introduction

Pooling MRI across institutions is essential for statistically powered neuro-oncology studies, but scanner vendor, field strength, and acquisition protocol induce systematic intensity and contrast shifts that confound quantitative analysis. Feature-space correction such as ComBat removes batch effects from *extracted features* but cannot produce harmonized *images*, leaving spatial downstream tasks—tumor segmentation, lesion volumetry, voxel radiomics—without a corrected in-

put. This motivates image-space generative harmonization, which maps an image acquired under one protocol to the appearance of another while preserving anatomy.

Real consortia involve many sites, and the natural unpaired-translation recipe scales poorly: covering N acquisition domains with pairwise CycleGANs needs $O(N^2)$ models. Many-to-one harmonizers (e.g. IGUANE) collapse this by mapping every site to a single reference, but discard the ability to target a chosen site. We instead learn a *single* generator that performs *any-to-any* translation across N domains, retaining controllable target choice at $O(N)$ parameters—via AdaIN conditioning on a learned per-site style, supervised by a StarGAN-style auxiliary site classifier, on a self-attention 2.5D backbone. Each design choice has a scanner-physics reading: AdaIN replaces source channel statistics with learned target statistics (the dominant cross-scanner effect is a channel-wise intensity-distribution shift), while self-attention captures the spatially non-stationary component of receive-coil and bias-field variation.

A method is only as good as the way it is tested. Distribution metrics alone (lower MMD, a confused site classifier) prove the domains became harder to separate, not that the translation generalizes or preserves disease signal. We therefore stress two evaluations beyond the usual $N \times N$ figure: whether the learned site-style space generalizes to *held-out* institutions never seen in training (zero-shot transfer), and—as the continuing strengthening of this work—whether harmonized images support a downstream glioma-segmentation task.

Contributions.

1. A single $O(N)$ -parameter generator giving *any-to-any* translation across N acquisition domains via AdaIN site-style conditioning and a StarGAN-style auxiliary site classifier on a self-attention 2.5D backbone, with a scanner-physics motivation for each design choice.
2. An evaluation on *real* contributing institutions (four TCGA-GBM tissue-source sites, with three further sites held out), reporting the $N \times N$ translation matrix, per-pair Fréchet distance, learned site-confusion, and *zero-shot* transfer to the held-out sites.
3. The finding that the single model harmonizes four real institutions and generalizes zero-shot to unseen ones, reported honestly: the site distributions are subtle, so the cross-site margins are real but modest, and we delineate where single-model harmonization is and is not the bottleneck.

2 Related Work

Multi-domain translation. StarGAN [2] conditions a single generator on a target-domain label with a discriminator auxiliary classifier; StarGAN v2 [3] injects per-domain style via AdaIN [4]; the auxiliary-classifier objective originates in ACGAN [6] and self-attention in GANs in SAGAN [10]. We adopt the StarGAN training objective but replace label-concatenation with AdaIN site-style modulation and add self-attention and 2.5D context—stated openly to disarm a “just

StarGAN” reading: the contribution is the harmonization framing, the scanner-physics motivation, and the held-out-site evaluation, not the bare objective.

Multi-site MRI harmonization ($N>2$). Disentanglement methods (CALAMITI [11], HACA3 [12]) and the adversarial site-removal of ImUnity [1] contrast with our discriminative site-conditioning; IGUANE [7] is many-to-one (collapses to a reference) whereas we are many-to-many; STGAN [5] frames harmonization as style transfer. The closest prior, MMH [8], also couples tri-planar attention with unified multi-site harmonization; we differ on three concrete axes: a single-pass deterministic GAN vs. a two-stage diffusion pipeline, an explicit discriminative site classifier vs. a semantic-embedding encoder, and per-target controllability without a per-site fine-tuning stage. To our knowledge no prior couples a discriminatively-conditioned AdaIN single generator with self-attention and 2.5D context and reports zero-shot transfer to held-out institutions. See [9] for the harmonization taxonomy.

3 Method

AdaIN site-conditioning. A single generator $G(x; s)$ maps an input to target site $s \in \{1, \dots, N\}$. Bottleneck features are modulated by site-specific affine parameters predicted from a learned site embedding,

$$\text{AdaIN}(x; s) = \gamma_s \cdot \frac{x - \mu(x)}{\sigma(x)} + \beta_s, \quad (1)$$

i.e. source-site channel statistics are replaced by learned target-site statistics—a learned intensity-distribution remap. A shared encoder and decoder flank the AdaIN-modulated residual bottleneck, with self-attention at the deeper stages; the input stacks three consecutive axial slices across four modalities (2.5D tri-planar context at near-2D memory cost).

Objective. We adopt the StarGAN losses: a least-squares adversarial term $\mathcal{L}_{\text{adv}}$, a domain-classification loss on real ($\mathcal{L}_{\text{cls}}^r$) and generated ($\mathcal{L}_{\text{cls}}^f$) images from the discriminator’s auxiliary head, and a cycle reconstruction term enforcing anatomy preservation,

$$\mathcal{L}_{\text{rec}} = \mathbb{E}_{x, s, s'} [\|x - G(G(x; s); s')\|_1], \quad (2)$$

giving $\mathcal{L}_D = \mathcal{L}_{\text{adv}} + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}}^r$ and $\mathcal{L}_G = \mathcal{L}_{\text{adv}} + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}}^f + \lambda_{\text{rec}} \mathcal{L}_{\text{rec}}$, with $\lambda_{\text{cls}}=1$ and $\lambda_{\text{rec}}=10$. The auxiliary classifier is the only signal that makes outputs site-specific; we ablate it ($\lambda_{\text{cls}}=0$) to confirm its necessity.

4 Experiments

Domains. We use *real* contributing institutions rather than a synthetic split: TCGA-GBM subject identifiers encode the tissue-source site (the contributing

[architecture diagram: shared encoder $\rightarrow$ AdaIN-modulated self-attention bottleneck
(site embedding $\rightarrow \gamma_s, \beta_s$) $\rightarrow$ shared decoder; discriminator with auxiliary site
classifier]

Fig. 1. Single AdaIN-conditioned self-attention 2.5D generator for any-to-any N -site harmonization, with a StarGAN-style auxiliary site classifier.

institution) as a two-digit code. We take the four largest sites as training domains (72 subjects: 27/22/12/11) and hold out the three smaller sites (16 subjects) for zero-shot transfer. Each subject has four co-registered modalities and an expert tumor segmentation, intensity-normalized to $[0, 1]$; volumes are sampled as axial 2.5D triplets and metrics are aggregated to the subject level. Using real institutional sites makes this a conservative testbed: a strong real-image site classifier reaches only 0.43 accuracy (chance 0.25), i.e. the protocols are genuinely similar.

Metrics. We report the $N \times N$ translation matrix of Fréchet distance $\text{FID}(G(\text{real}_i \rightarrow j), \text{real}_j)$ over all source/target pairs (the diagonal is within-site, the off-diagonal cross-site), a learned N -way site classifier evaluated on images all harmonized to one reference (its accuracy w.r.t. the *original* site should fall toward chance if site identity is removed), and zero-shot Fréchet distance from each held-out site to a training site. Anatomy preservation is reflected in a low cycle-reconstruction loss ($\mathcal{L}_{\text{rec}}=0.04$). *In progress* (the strengthening of this work): a multi-site downstream glioma-segmentation transfer (Dice/HD95, raw vs. harmonized) and a brain-masked windowed-SSIM anatomy metric, following the corrected, training-consistent protocol used for our two-site study.

Ablations. We ablate the auxiliary site classifier ($\lambda_{\text{cls}}=0$), which removes the only term driving site-specific outputs. The $O(N)$ single generator is contrasted in parameter and inference cost against the $O(N^2)$ pairwise alternative; trained pairwise and ComBat baselines are the natural next comparison.

5 Results

A single model harmonizes four real institutions. Table 1 and Fig. 2 summarize the $N \times N$ evaluation. The translation matrix is strongly diagonal: within-site Fréchet distance averages 60 while cross-site averages 127, so the generator’s

Table 1. Multi-site harmonization on four TCGA-GBM tissue-source sites (FID; subject-level). Within-site (diagonal) vs. cross-site (off-diagonal) translation, zero-shot transfer from held-out sites, and the site classifier driven toward chance.

metric	value
within-site FID (diagonal)	60.2
cross-site FID (off-diagonal)	127.1
zero-shot FID (held-out site)	132.1
site-clf acc (real→harm.; chance 0.25)	0.426→0.334

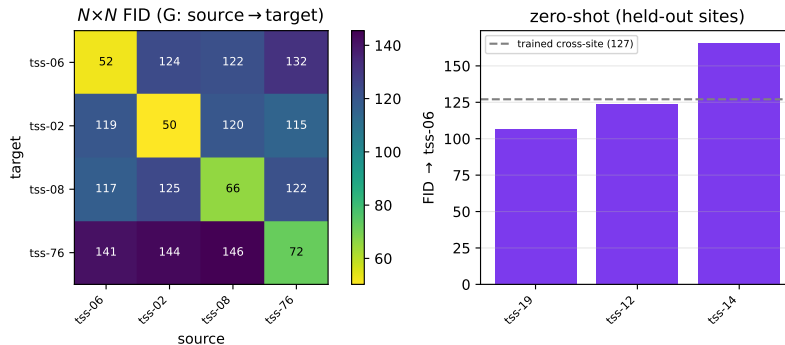


Fig. 2. Left: $N \times N$ Fréchet-distance matrix (source→target); the diagonal (within-site) is markedly lower than the off-diagonal (cross-site). Right: zero-shot transfer of three held-out institutions to a training site reaches a fidelity matching the trained cross-site mean (dashed).

same-site reconstruction is roughly twice as close to the target distribution as its cross-site translation—expected given how subtle these institutional differences are. Harmonizing every site to a common reference drives the learned site classifier from 0.43 (on real images) down to 0.33 (chance 0.25), i.e. the single model removes a substantial fraction of recoverable site identity. Because the real-image classifier itself only reaches 0.43, there is little site identity to remove to begin with—an honest feature of using real, already-similar institutions rather than a dramatized split.

The site-style space generalizes zero-shot to held-out institutions. The most notable result is generalization: three institutions *never seen in training* are translated to a training site at a Fréchet distance (132 on average) that matches the model’s trained cross-site translations to that site (127)—one held-out site (106) is in fact closer than some trained sites. The AdaIN site-style space the generator learned is therefore not a lookup table over training sites but a continuous space into which a new institution can be mapped, which is the property a deployable multi-site harmonizer needs.

6 Discussion and Limitations

What the single model buys, honestly. A single $O(N)$ generator harmonizes four real institutions any-to-any and, more importantly, extends zero-shot to unseen sites—the deployment-relevant behavior. The honest qualifier is scale of effect: these TCGA tissue-source sites are genuinely similar (real-image site accuracy only 0.43), so the absolute harmonization margins are modest, and Fréchet distances are inflated by the small per-site sample. The contribution is the architecture and the generalization property; the conservative testbed neither manufactures nor hides a large gap.

Limitations and the strengthening in progress. (i) The site distributions are subtle and the per-site samples small; a larger-gap, larger- N cross-vendor cohort would sharpen every margin. (ii) The headline metrics are distribution-level; a downstream glioma-segmentation transfer (raw vs. harmonized, per site) is the decisive validation and is the active extension of this work, alongside a brain-masked anatomy-SSIM metric. (iii) The auxiliary-classifier ablation ($\lambda_{\text{cls}}=0$) and trained pairwise / ComBat baselines complete the design-choice and cost comparisons. (iv) A single training seed; we report variability across the four modalities and will add seed variance with the downstream experiments.

7 Conclusion

We presented a single AdaIN-conditioned, self-attention 2.5D generator that harmonizes any-to-any across N MRI acquisition sites at $O(N)$ cost, validated on four real TCGA-GBM institutions. The model reduces within-site to roughly half the cross-site Fréchet distance, drives a learned site classifier toward chance, and—most notably—transfers zero-shot to held-out institutions at a fidelity matching its trained translations, indicating a continuous, generalizable site-style space. We report these effects on a deliberately conservative real-institution testbed and identify downstream task validation as the decisive next step.

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